

Lab 4

● Graded

Group

Kyle Connall

Fady Youssef

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Total Points

34 / 35 pts

Question 1

Exercise 1

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Small error in sample size. Answer: $n=385$
- 2 pts Sample size for part b is incorrect. Answer: $n=385$

Question 2

Exercise 2

4 / 4 pts

✓ - 0 pts Correct

- 1 pt The confidence interval is incorrect. Answer: (0.539, 0.600)
- 1.5 pts The confidence interval is not correctly interpreted in the context of the problem.
- 2 pts Interpretation for the confidence interval is missing

Question 3

Exercise 3

4 / 4 pts

✓ - 0 pts Correct

- 0.5 pts Alternative hypothesis is incorrect.
- 0.5 pts Mathematical notation for null and alternative hypotheses is incorrect
- 1 pt Null and alternative hypotheses are incorrect or missing
- 1 pt p-value is incorrect
- 1 pt Missing or Error in the conclusion to the hypothesis test
- 2 pts The conclusion is incorrect and not stated in the context of the problem.
- 4 pts Missing answer and supporting work

Question 4

Exercise 4

4 / 5 pts

✓ - 0 pts Correct

- 1 pt The sample proportion is incorrect.

- 1 pt p-value is incorrect

✓ - 1 pt The conclusion to the hypothesis test is incorrect

- 2 pts The conclusion is incorrect and not stated in the context of the problem.

Question 5

Exercise 5

5 / 5 pts

✓ - 0 pts Correct

- 2 pts The sample proportion is incorrect. Answer: $(342+143)/1392$.

Question 6

Exercise 6

5 / 5 pts

✓ - 0 pts Correct

- 2 pts The sample proportion is incorrect. Answer: $67/118$

Question 7

Exercise 7

4 / 4 pts

✓ - 0 pts Correct

- 1 pt Error in the confidence interval. Answer: $(0.119, 0.181)$

- 0 pts The interpretation of the confidence interval is incorrect or missing.

Question 8

Exercise 8

4 / 4 pts

✓ - 0 pts Complete. Graded on completeness.

- 4 pts Problem is missing

Kyle Connall
Fady Youssef
04/14/2025

Lab 4: Hypothesis Testing & Confidence Intervals for Proportions

Exercise 1: For each of the following sample sizes, find the margin of error

- a. If Gallup picks a sample size of 400, what would the margin of error be for a 95% confidence interval?

```
#1
# Given values
phat = 0.5
n = 400
z = 1.96

#A.
# MOE
MOE = z * sqrt(phat * (1 - phat) / n)
#MOE = 0.049
```

Margin of error = .049 = 4.9%

- b. Which sample size is best if Gallup wants to do the least expensive poll (the smallest possible sample) that gives a margin of error below 5%?

```
#B.
targetMOE = 0.05

# Solve for n
min_n = (phat * (1 - phat)) * (z / targetMOE)^2
ceiling(min_n)
#min_n = 385
|
```

385 respondents required to maintain a margin of error below 5%

Exercise 2: The title of the article in question was “*Americans’ Desire for Third Party Persists This Election Year*”, and the first sentence states that 57% of Americans say that a third major U.S. political party is needed. The “57%” is a point estimate for the actual feelings of the entire United States. Use sample size $n = 1020$.

- a. What is the standard error of this estimate?

```
#2
#Given
phat2 = 0.57
n2 = 1020

#A.
SE2 = sqrt(phat2 * (1 - phat2) / n2)
#SE2 = 0.01550142
```

The standard error is about 0.0155.

- b. What is the actual margin of error for this proportion?

```
#B.
z2 = 1.96
MOE2 = z2 * SE
#MOE2 = 0.03038279
```

The margin of error is about 0.0304, meaning we expect the true population proportion to fall within 3.04 percentage points of our sample estimate.

- c. Create a 95% confidence interval around the 57% statistic that gives a range estimate for the proportion of US adults who think a third major U.S. political party is needed. Then write a sentence explaining the meaning of “95% confidence” in this context.

```
#C
lower = phat2 - MOE2
upper = phat2 + MOE2
c(lower, upper) #= 0.5396172 0.6003828
```

we are 95% confident that the true proportion of U.S. adults who think a third major political party is needed falls between 53.96% and 60.04%.

The article states that in past election years (2008 and 2012) “...Americans were evenly divided on the need for a third party.” Does this more recent poll with 57% in favor represent a shift in opinion, or just natural variation in the random sample? If our poll had said 51% were in favor, we would be unlikely to say this represents a shift in opinion. But is 57% a statistically significant change? To determine this, you will be asked to conduct a hypothesis test.

Steps for a Hypothesis Test for a 1-Proportion Scenario

- Step 0: Read the problem. What is the story? What is being investigated?
- Step 1: State the null and alternative hypotheses.
- Step 2: Find the sample data: n = sample size, $\hat{p} = \frac{x}{n}$
- Step 3: Find the mean and standard error: Mean = p & $SE = \sqrt{\frac{p \cdot (1-p)}{n}}$
- Step 4: Find the p-value (use pnorm()). Find the test-statistic (z-score), $z = (\hat{p} - p)/SE$.
- Step 5: Interpret the p-value and state your conclusions. Provide your conclusions using complete sentences and relating back to the specific context. Follow up with a confidence interval if needed.

Exercise 3: Conduct a full hypothesis test to determine if the proportion of people in favor of a 3rd party has increased from the previous 50%. Use the sample data $n=1020$, $\hat{p} = 0.57$ and a significance level of $\alpha = .05$. Provide all relevant statistics in your lab write-up, including the standard error and p-value. While you may state “reject” or “fail to reject” in your conclusions, also phrase your conclusions in everyday language related to the context of this opinion poll.

```
#3
#Given
p03 = 0.50
phat3 = 0.57
n3 = 1020

SE3 = sqrt(p0 * (1 - p03) / n3)

z3 = (phat3 - p03) / SE3
pvalue3 = 2 * (1 - pnorm(z3))
#z3 = 4.471241
#pvalue3 = 7.776684e-06
```

Conclusion: Since the p-value is much smaller than 0.05, we reject the null hypothesis. This means there is strong evidence that support for a third major U.S. political party has increased from 50% to 57% indicating more Americans now believe a third party is needed compared to previous years.

We will revise data from the 2012 US Census American Community Survey in the data set: acs12. This is a sample of data from 2000 people. There are 13 columns of data. We will focus on categorical data, including:

- “employment”, which gives employment status as “not in labor force”, “unemployed”, “employed”, or NA.
- “race”
- “gender”
- “edu” – education level, categorized as high school or lower, college, graduate, or NA.
- “disability”

For each of the problems 4-6, conduct a hypothesis test using of the steps listed in the box on page 2. For each problem, assume the hypothesis is a two-tailed test because we don’t know if the alternate should be greater than or less than the null hypothesis. Also assume $\alpha = .05$

Exercise 4: A researcher is curious about gender differences and employment. One could assume that half of the employed people are men and half are women. Conduct a hypothesis test to determine whether the data in the 2012 American Community Survey supports this hypothesis. Be sure to look only at the employed respondents in this data set. Since gender and employment are categorical variables, we can use a table to summarize the data (see Lab 2).

```
#4
library(openintro)
# sample values
employed = subset(acs12, employment == "employed")

table(employed$gender)
# male | female
# 470   373

x4 = 373
n4 = 843
phat4 = x4 / n4
p0 = 0.5

SE4 = sqrt(p0 * (1 - p0) / n4)

z4 = (phat4 - p0) / SE4

pvalue4 = 2 * (1 - pnorm(z4))

cat("Z-score:", z4, "\n")
cat("P-value:", pvalue4, "\n")
#Z-score: -3.340857
#P-value: 1.999165 |
```

Conclusion: Since the p-value is much bigger than 0.05, we don’t have enough evidence to say there is a difference. So, the data still supports the idea that about half of employed people are men and half are women.

Exercise 5: A report from 2000 indicated that 31% of adults aged 24 or older had a college degree. We are curious if this rate is still accurate for 2012. The American Community Survey 2012 data (acs12) includes 3 levels of education: High school and lower, college, and graduate school. When we say college degree, we would want to include both college and graduate school. Conduct a hypothesis test to determine if the proportion of 31% of adults is consistent with the acs12 data.

Note: In order to investigate this claim using the acs12 data set, you will need to create a subset of the data for people ages 24 and up (hint: there are 1392 people in this age range). Creating a table will not work because age is not a categorical variable. For a review of the subset command, you can revisit Lab 3, or check your R cheat sheet.

```
#5
adults = subset(acs12, age >= 24)
table(adults$edu)

#hs or lower    college    grad
#      907         342      143

#H0:p = 0.31
#HA:P != 0.31

# Inputs
x5 = 485
n5 = 1392
phat5 = x5 / n5
p05 = 0.31

# Standard error
SE5 = sqrt(p05 * (1 - p05) / n5)

# Z-score
z5 = (phat5 - p05) / SE5

# Two-tailed p-value
p_value5 = 2 * (1 - pnorm(z5))

# Output
cat("Sample proportion:", round(phat5, 4), "\n")
cat("Z-score:", round(z5, 4), "\n")
cat("P-value:", round(p_value5, 4), "\n")
#Sample proportion: 0.3484
#Z-score: 3.0993
#P-value: 0.0019 |
```

Conclusion: Since the p-value is 0.0019, which is much smaller than 0.05, we reject the null hypothesis. This means the proportion of adults aged 24 or older with a college degree in 2012 is significantly different from the 31% reported in 2000.

Exercise 6: A researcher is curious about gender differences among non-citizens. It would be reasonable to think that half of non-citizens are men and half are women. Conduct a hypothesis test to determine whether the data in the 2012 US Census American Community Survey supports this hypothesis. Be sure to look only at the non-citizens in this data set, which greatly reduces your sample size.

#6

```
#H0:p = 0.5
#HA:P != 0.5

non_citizens = subset(acs12, citizen == "no")
gender_table = table(non_citizens$gender)
print(gender_table)
#male female
#51      67

x6 = sum(non_citizens$gender == "male")
n6 = 103
p_hat6 = x6 / n6
p06 = 0.5
SE6 = sqrt(p06 * (1 - p06) / n6)
z6 = (p_hat6 - p06) / SE6
#z6 = -0.09853293
p_value6 = 2 * (1 - pnorm(z6))
#p_value6 = 1.078491
```

Conclusion: Since the p-value is bigger than 0.05, we don't have enough evidence to say the number of male and female non-citizens is different. So, the data does not show a real gender difference among non-citizens.

In the final section, there are no hypotheses to test. Instead, we want to find confidence intervals. Confidence intervals are often used in scenarios where we don't have pre-existing theories.

Russians' Opinions on Russian Influence:

There has been a great deal of reporting on Russian interference in American elections. But what do the citizens of Russia think of this? The following data set, contained in the OpenIntro library: `russian_influence_on_us_election_2016` contains survey responses from 506 Russian citizens who were asked whether their government attempted to influence the 2016 election in the United States. Their responses were "Did not try", "Did try", and "Don't know."

Exercise 7: Create a 95% confidence interval for the proportion of Russian citizens who agree with the claim that their government did try to influence the US election.

```
#7
data(russian_influence_on_us_election_2016)
table(russian_influence_on_us_election_2016$influence_2016)

#Did not try | Did try | Don't know
#          359 |       76 |         71

x7 = 76
n7 = 506

phat7 = 76/359
#phat7= 0.2116992
SE7 = sqrt((.2117*(1-.2117))/506)

z = 1.96
MOE7 = z * SE7
#MOE7 =0.03559484
```

Conclusion: A 95% confidence interval for the proportion of Russian citizens who said their government did try to influence the 2016 U.S. election is about 0.1761 to 0.2477. This means we are 95% confident that between 17.6% and 24.8% of Russian citizens believe their government was involved.

Exercise 8: Create a 99% confidence interval for the same scenario as Exercise 7. Then explain why the 99% confidence interval should be wider than the 95% confidence interval.

#8

```
x8 = 76
n8 = 506

phat8 = 76/359
phat8
SE8 = sqrt((.2117*(1-.2117))/506)

z8 = qnorm(.995)
#z8 = 2.575829

MOE8 = z8* SE8
#MOE8 = 0.04677869
```

Conclusion: The 99% confidence interval is wider than the 95% one because it gives us more certainty that the true proportion is in the range. To be more confident, we need to include more possible values, which makes the interval larger. This tradeoff means we gain confidence but lose some precision.